THE UNIVERISITY OF DA NANG
UNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty of Advanced Science and Technology

SOCIALIS REBUBLIC OF VIETNAM
Independence - Freedom - Happiness

UNDERGRADUATE PROGRAM

Final award: Degree of Bachelor/ Engineering

Programme title: Program Code:

COURSE SYLLABUS

Course title: Power Electronics

1. Course title:	Power Electronics
2. Course code:	
3. Credits:	02
4. Course plan	
- Lecture:	1.5 credits (22.5 hours)
- Tutorial:	… credits (... hours)
- Laboratory:	0.5 credits (15 hours)
- Self-learning:	60 hours
5. Coordinator and committees:	
- Coordinator:	Nguyen Le Hoa, Ph.D.
- Committee:	Le Quoc Huy, Ph.D., Giap Quang Huy, Ph.D.
- Teaching faculty:	Faculty of Advanced Science and Technology
6. Prerequisite requirements:	
- **Enforced prerequisites**:	N/A
- **Prerequisites**:	Electric Circuits 2, Electronics Devices and Circuits 1
- **Corequisites**:	N/A
7. Course type:	☐ Required ☒ Selected Elective ☐ Free elective
8. Knowledge Cluster	☐ Math and basic science ☐ General education ☒ Major education ☐ Internship/Capstone Project

9. Course description:

This course is a major specialization course providing fundamentals of power electronics. The contents of this course include: Introduction to power electronics; overview of power semiconductor switches; rectifier circuits that includes uncontrolled

rectifiers (diode rectifiers) and controlled rectifiers (thyristor rectifiers); DC-DC converters (Buck converter, Boost converter, Buck-Boost converter, Cúk converter,..); DC-AC inverters (single phase inverters, three-phase inverters); and applications of power electronics. Also, through this course, Matlab/Simulink will be use to simulate power electronic converters and inverters.

10. Course learning outcomes (CLOs):

At the end of this course, students should be able to:

No	(CLOs) (1)	Knowledge (2)	Skills (3)	Attitudes (4)	PLOs
1.	**Analyze** the working principles of the most common power semiconductor switches (diodes, thyristor, BJT, MOSFET, IGBT)	L4-Analysing	L3-Precision	L2-Responding	PLO1
2.	**Analyze** the characteristics of rectifier circuits (uncontrolled rectifier, controlled rectifiers) and compare the difference between rectifier circuits	L4-Analysing	L3-Precision	L2-Responding	PLO8
3.	**Analyze** the characteristics of DC-DC converters and DC-AC inverters	L4-Analysing	L3-Precision	L2-Responding	PLO8
4.	**Design** power electronic converter/inverter that meet the requirements of particular applications	L6-Creation	L3-Precision	L3-Valuing	PLO2
5.	**Use** Matlab/Simulink to simulate power electronic converter/inverter and expolore the responses of the converter/inverter	L3 - Applying	L3-Precision	L2-Responding	PLO6

11. Outcome Coverage: mapping to Program Learning Outcome (PLO)

PLOs	PLO1	PLO2	PLO3	PLO 4	PLO 5	PLO6	PLO 7	PLO8	PLO9
Course contribution	T1.1.3 T1.2.3 T1.3.3	T2.1.3 T2.2.3		I4.1 I4.1 I4.3	U5.1.3 U5.2.3 U5.3.3 U5.4.3 U5.4.4	T6.1.3 T6.2.3 T6.3.3 T6.4.3		T8.1.4 T8.2.3 T8.3.3	U9.1.2 U9.2.2 U9.3.2 U9.4.2 U9.5 U9.6 U9.9 U9.10
CLO 1	X								

CLO 2								X	
CLO 3								X	
CLO4		X							
CLO5						X			

12. Student Responsibilities:

Students must:

- Attend at least 80% of the course to be eligible for the final examination
- Engage in class discussion with respect and attention
- Self-study, direct their own studying—outside the classroom
- Complete all homework and assignments in a timely manner

13. Course assessment:

Assessment Components	Assessment types	Rubric	Percentages (%)	Percentages of assessment components (%)	CLOs
A – On-going Assessment	A1. Quiz	R1.1 – rubric of PI 1.1 R1.2 – rubric of PI 1.2 R1.2 – rubric of PI 1.2 R2.1 – rubric of PI 2.1 R2.2 – rubric of PI 2.2 R8.1 – rubric of PI 8.1 R8.2 – rubric of PI 8.2	10	20	CLO1 CLO2 CLO3 CLO4

	A2. Weekly homework	R1.1 – rubric of PI 1.1 R1.2 – rubric of PI 1.2 R1.2 – rubric of PI 1.2 R2.1 – rubric of PI 2.1 R2.2 – rubric of PI 2.2 R8.1 – rubric of PI 8.1 R8.2 – rubric of PI 8.2	10		
B – Midterm exam	B1. Written test	R1.1 – rubric of PI 1.1 R1.2 – rubric of PI 1.2 R1.2 – rubric of PI 1.2 R2.1 – rubric of PI 2.1 R2.2 – rubric of PI 2.2 R8.1 – rubric of PI 8.1 R8.2 – rubric of PI 8.2	20	20	CLO1 CLO2 CLO3
C-Final exam	C1. Written test	R1.1 – rubric of PI 1.1 R1.2 – rubric of PI 1.2 R1.2 – rubric of PI 1.2 R2.1 – rubric of PI 2.1 R2.2 – rubric of PI 2.2 R8.1 – rubric of PI 8.1 R8.2 – rubric of PI 8.2	40	40	CLO1 CLO2 CLO3 CLO4

D-Laboratory	D1. Conductions of experiment	R6.1-rubric of PI 6.1	5	20	CLO5
	D2. Data acquisition	R6.2-rubric of PI 6.2	5		
	D3. Data evaluation	R6.3-rubric of PI6.3	5		
	D4. Presentation with report	R3.4-rubric of PI3.4	5		

14. Lecture plan

14.1. Lecture plans and activities

Weeks (hours)	Contents	Teaching and learning activities	Assessment Types	CLOs
1 (2)	Chapter 1. Power electronic systems 1.1 Introduction 1.2 Power electronics versus linear electronics 1.3 Scope and applications 1.4 Classification of power processors and converters	**Teaching:** - Introduction to the course syllabus (course type, knowledge cluster, prerequisite, course description, CLOs, mapping to PLOs, student responsibility, course assessment, lecture plan, textbook and references, scientific ethics,..) - Deliver the lecture and ask students questions **In-class study:** - Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 1	A1, A2	CLO1
2 (2)	Chapter 2. Review of power semiconductor switches 2.1 Diodes 2.2 Thyristors	**Teaching:** - Deliver the lecture - Ask students questions - Give a quiz **In-class study:**	A2	CLO1

	2.3 Bipolar junction transistors 2.4 Metal-oxide-semiconductor field effect transistors 2.5 Isolate gate bipolar transistors	- Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions - Take a quiz **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 2		
3 (2)	Chapter 3. Rectifiers 3.1 Basic rectifier concepts 3.2 Diode rectifiers (single phase, three-phase, full bridge,..)	**Teaching:** - Deliver the lecture - Ask students questions **In-class study:** - Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 3	A2	CLO2
4 (2)	Chapter 3. (cont'd) 3.3 Thyristor circuits and their control 3.4 Single phase thyristor rectifiers	**Teaching:** - Deliver the lecture - Ask students questions - Give a quiz **In-class study:** - Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions - Take a quiz **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 3	A1, A2	CLO2
5(2)	Chapter 3. (cont'd) 3.5 Three-phase thyristor rectifiers	**Teaching:** - Deliver the lecture - Ask students questions	A1, A2	CLO2

		- Give a quiz **In-class study:** - Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions - Take a quiz **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 3		
6 (5)	Laboratory Simulate rectifiers using Matlab/Simulink		D1, D2, D3, D4	CLO5
7 (2)	Chapter 4. DC-DC switch mode converters 4.1 Control of DC-DC converters 4.2 Step-down (Buck) converter 4.3 Step-up (Boost) converter	**Teaching:** - Deliver the lecture - Ask students questions **In-class study:** - Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 4	A2	CLO3 CLO4
8	Midterm exam		B1	CLO1 CLO2 CLO3 CLO4
9 (2)	Chapter 4. (cont'd) 4.4 Buck-Boost converter 4.5 Cúk converter 4.6 DC-DC converter comparison	**Teaching:** - Deliver the lecture - Ask students questions **In-class study:** - Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions	A2	CLO3 CLO4

		Self study: - Read textbooks and other relevant materials - Do homework of Chapter 4		
10 (5)	Laboratory Simulate DC-DC converters using Matlab/Simulink		D1, D2, D3, D4	CLO5
11 (2)	Chapter 5. Switch mode DC-AC inverters 5.1 Introduction 5.2 Basic concepts of switch-mode inverters	**Teaching:** - Deliver the lecture - Ask students questions - Give a quiz **In-class study:** - Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions - Take a quiz **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 5	A1, A2	CLO3 CLO4
12 (2)	Chapter 5. (cont'd) 5.3 Single phase inverters	**Teaching:** - Deliver the lecture - Ask students questions - Give a quiz **In-class study:** - Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions - Take a quiz **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 5	A1	CLO3 CLO4
13 (2)	Chapter 5. (cont'd) 5.4 Three-phase inverters	**Teaching:** - Deliver the lecture - Ask students questions - Give a quiz **In-class study:**	A1	CLO3 CLO4

		- Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions - Take a quiz **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 5		
14 (5)	Laboratory Simulate DC-AC converters using Matlab/Simulink		D1, D2, D3, D4	CLO5
15 (2,5)	Chapter 6. Applications of power electronics 6.1 Switching DC power supplies 6.2 Power conditioners and uninterruptible power supplies	**Teaching:** - Deliver the lecture - Ask students questions **In-class study:** - Attend the lecture and listen - Take your own notes on main points - Respond to questions raised by lecturer - Ask questions **Self study:** - Read textbooks and other relevant materials - Do homework of Chapter 6	A1	CLO3 CLO4
16	Final exam		C1	CLO1 CLO2 CLO3 CLO4

14.2. Lab plans and activities

Weeks	Contents	Teaching and learning activities	Assessment Types	CLOs
6	**Lab 1**: Simulate rectifiers using Matlab/Simulink	**Teaching**: - Review theory - Introduction to control toolbox of Matlab - Introduction to content of the lab	D1, D2, D3, D4	CLO5

		In-class study: - Use Matlab to simulate the assigned problems - Record data and analysis		
10	**Lab 2**: Simulate DC-DC converters using Matlab/Simulink	**Teaching**: - Review theory - Introduction to control toolbox of Matlab - Introduction to content of the lab **In-class study**: - Use Matlab to simulate the assigned problems - Record data and analysis	D1, D2, D3, D4	CLO5
14	**Lab 3**: Simulate DC-AC inverters using Matlab/Simulink	**Teaching**: - Review theory - Introduction to control toolbox of Matlab - Introduction to content of the lab **In-class study**: - Use Matlab to simulate the assigned problems - Record data and analysis	D1, D2, D3, D4	CLO5

15. References:

15.1 Textbooks, course books:

[1]. [NED et al., 2003] Power Electronics – Converters, Applications, and Design, 3rd Ed., John Willey & Son, Inc.

15.2 Reference books:

[1]. [MUHAMMAD, 2014] Power Electronics – Devices, Circuits, and Applications, 4th Ed., Pearson.

16. Scientific ethics:

- Students must respect their lecturers and other students.
- Students must comply with the university's academic integrity.
- Students must strictly follow the rules and regulations of the university.

17. Approval date:

18. Approval level:

Dean of Faculty	**Program chair**	**Lecturer in charge**
TS. Nguyễn Lê Hòa		TS. Nguyễn Lê Hòa